GENERAL INFORMATION

00 SECTION

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GENERAL INFORMATION . . . 00-00

00-00 GENERAL INFORMATION

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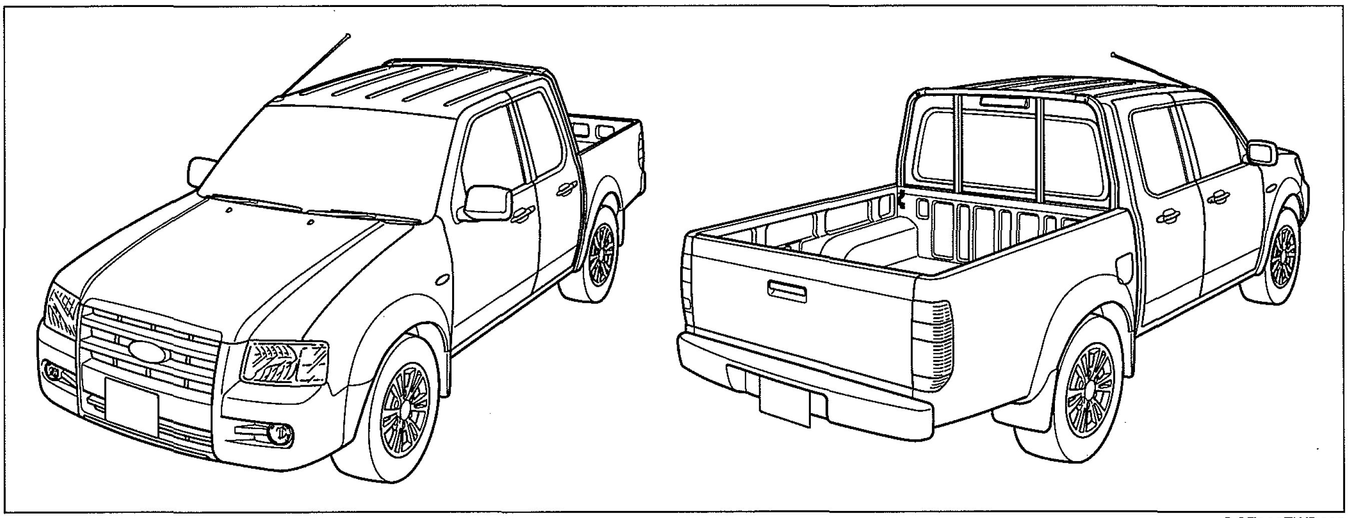
AIM OF DEVELOPMENT

dcf000000000t01

External View



DCF000ZWB002

Vehicle Outline

Engine

Mechanical

- A roller has been adopted to the rocker arm to reduce rolling resistance. Due to this, fuel economy has been improved. (WL-C, WE-C)
- Valve lift has been increased (high lift), improving intake and exhaust efficiencies. Due to this, engine torque and power have been improved. (WL-C, WE-C)
- A timing belt auto tensioner has been adopted to reduce timing belt deterioration. (WL-C, WE-C)
- A DOHC type valve system has been adopted for improved engine torque and power. (WL-C, WE-C)
- Due to the bifurcated oil pan bottom, vibration in the oil pan has been reduced for improved engine noise suppression. (WL-C, WE-C)
- A hard-plastic oil strainer has been adopted for weight reduction.
- A cover has been installed on the upper part of the engine for improved appearance and engine noise suppression. (WL-C, WE-C)
- A balance shaft has been adopted for reduced vibration and noise. (G

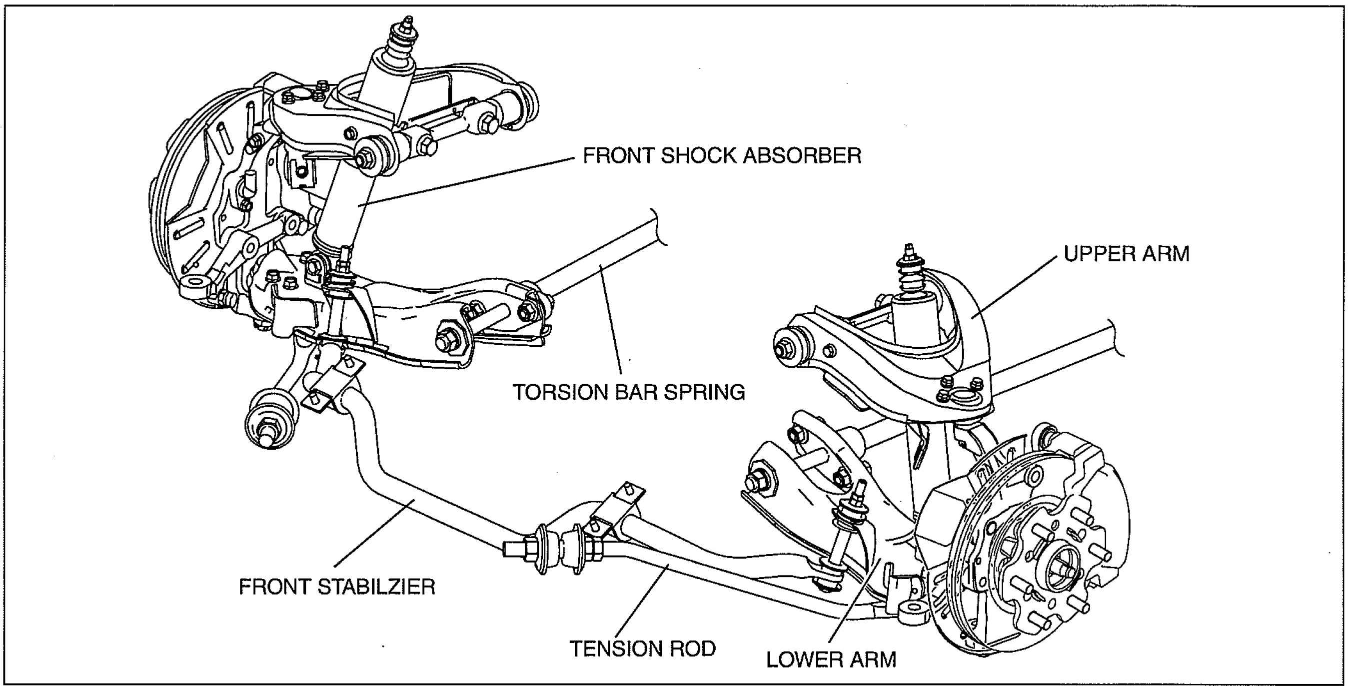
Intake/exhaust and fuel supply systems

- Superior high-output, noise suppression, and low fuel consumption have been realized due to the adoption of a direct injection type common rail. (WL-C, WE-C)
- Smooth driveability has been realized due to the adoption of a variable induction turbocharger which achieves a balance between the air charging pressure providing responsive rise from low speed and the air charging pressure at the high speed range. (WL-C, WE-C)
- A water-cooling type EGR cooler with excellent cooling capability has been adopted for improved intake air charging efficiency, recirculating more exhaust gas. Due to this, NOx has been reduced for improved exhaust emission performance.

Suspension and steering

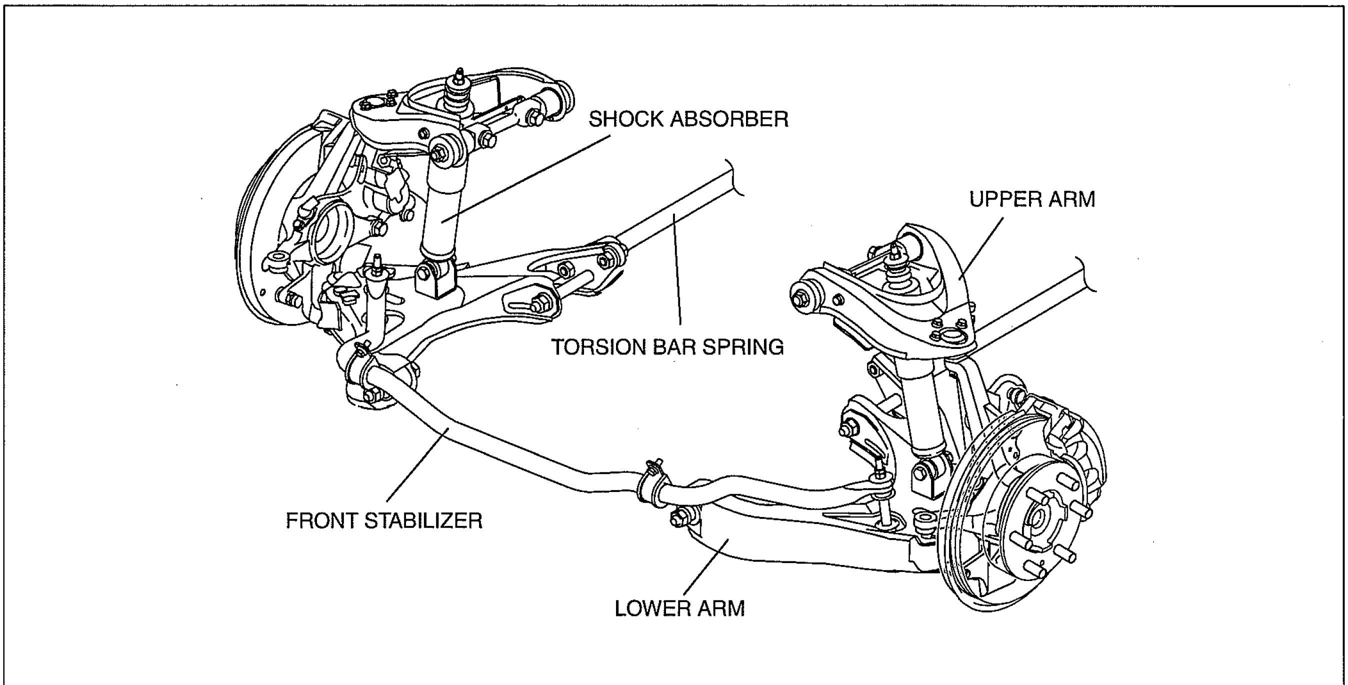
- Front suspension
  — The double wishbone suspension, which has I-shape lower arms and tension rods, is used for the front suspension of 4x2 models.
  — The double wishbone suspension, which has A-shape lower arms, is used for the front suspension of 4x2 (Hi-Rider) and 4x4 models. A-shape lower arms have high rigidly and durability.
  — The torsion bar springs are used for both 4x2 and 4x4 models.

4x2



DBG213ZTB001

4x2 (Hi-Rider)



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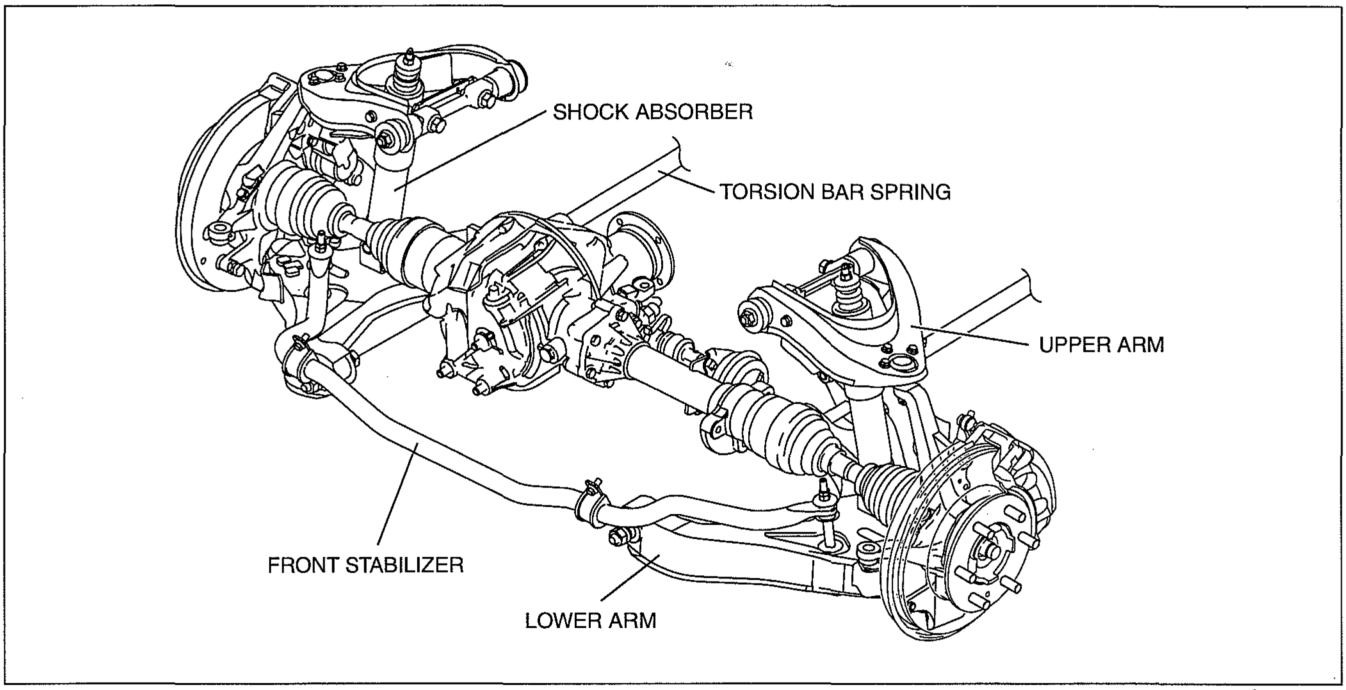
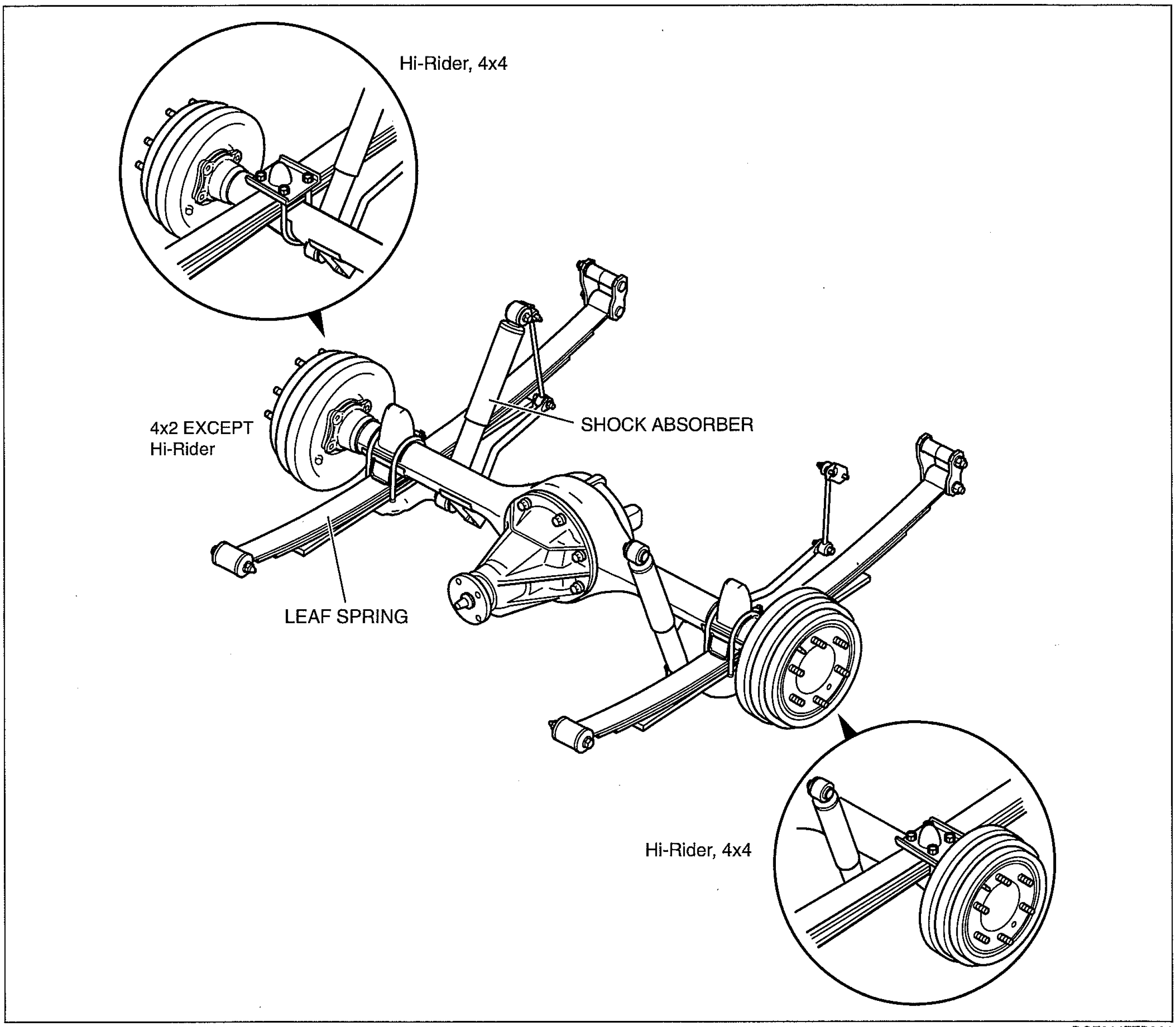
4x4



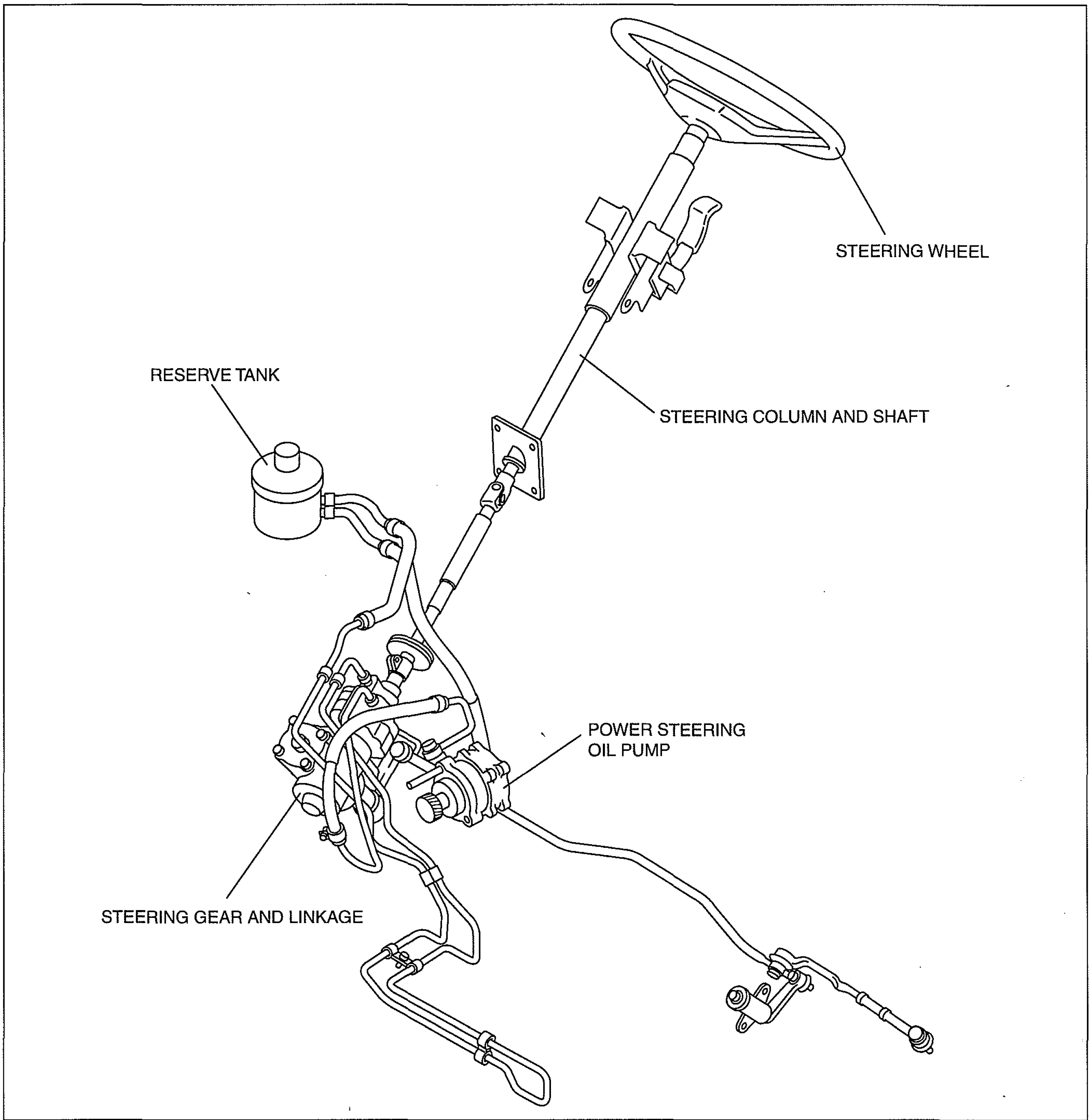
DBG213ZTB003

- Rear suspension
  — The rigid axle suspension with leaf spring is used for the rear suspension.
  — To prevent wind-up of the leaf springs during rapid acceleration or deceleration, the shock absorber on the right side is mounted at the rear of the axle and the left one is mounted at the front of the axle, (bias mount).In order to increase the rigidity in the lateral direction, wide shackle plates are used in the frame mounting.
  — For 4x4 models, the rear springs are mounted on top of the axle housing to raise the center of gravity. Thus, the shock absorbers attach directly to brackets welded to the axle housing.



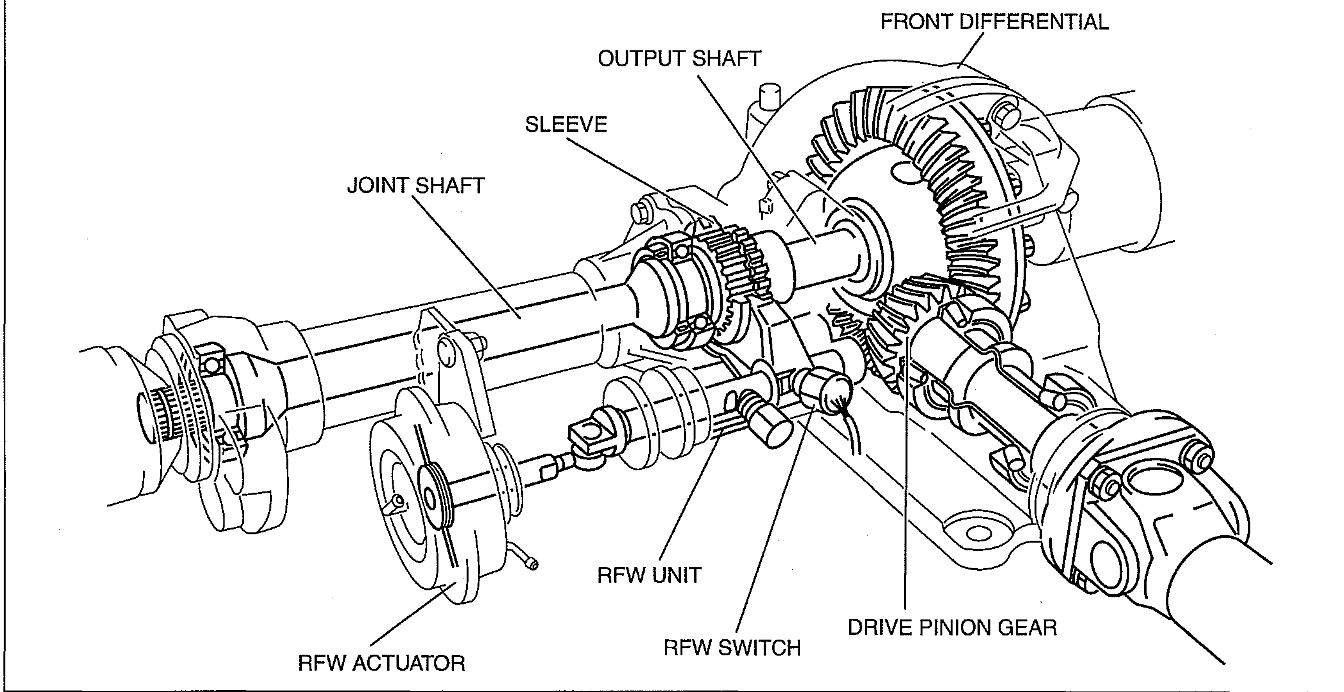
DCF214ZTB001

- Power steering
  — With the adoption of an engine speed sensing power steering mechanism, handling stability has been improved.
  — With the adoption, for all vehicles, of a steering column with a tilt mechanism, operability has been improved. (Some models)
  — With the adoption of a steering shaft with an energy absorbing mechanism, safety has been improved.



DBR614ZWB801

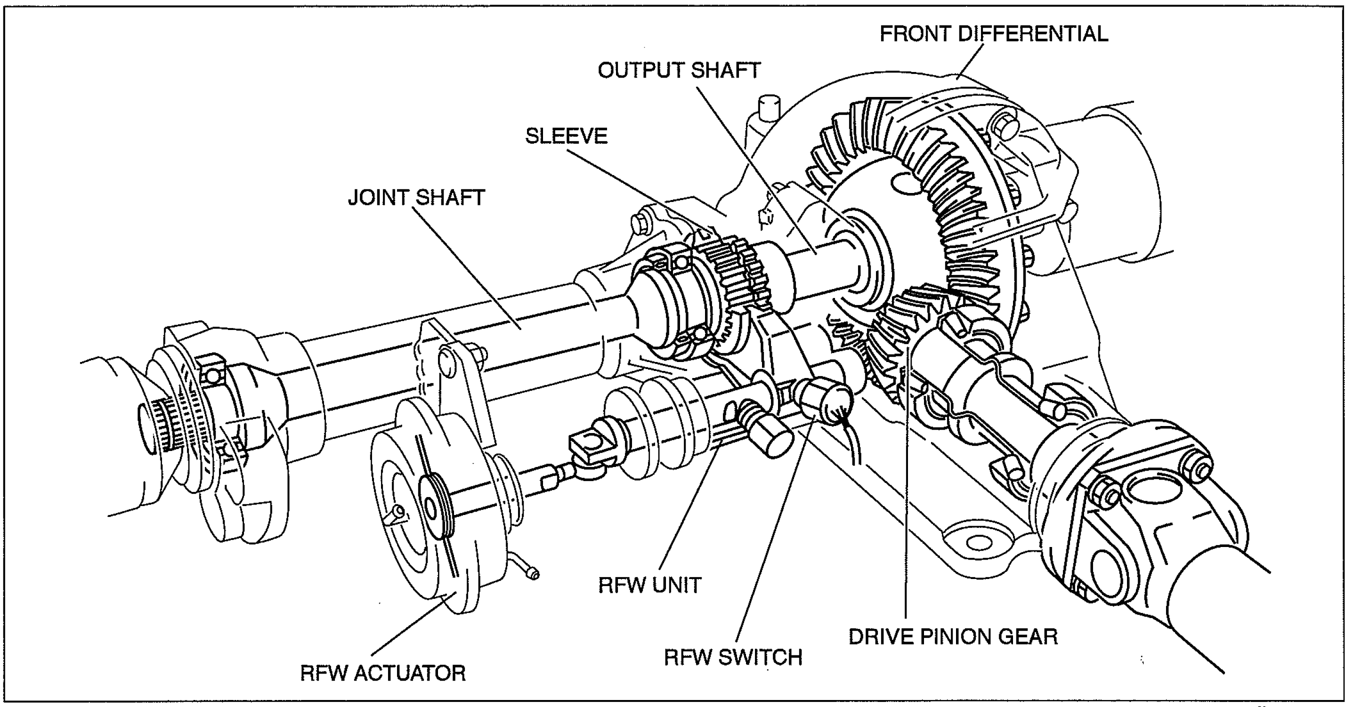
- Remote Freewheel (RFW)
  - — The PCM receives signals from the transfer case switch and other switches. It then activates one of two solenoids, causing a vacuum from the vacuum pump to be applied to the RFW actuator, and switching the remote freewheel (RFW) unit to lock or free.
  - — To prevent reduced performance of the actuator as a result of a decrease in engine vacuum or vacuum pump, such as during acceleration or at high altitude operation, the actuation system includes a one-way check valve.
  - — The switching mechanism of the RFW unit is a mechanical dog clutch on the left side of the front differential.
  - — The RFW operation is controlled by PCM.



DBG314ZTB012

4x4 control

- Remote Freewheel (RFW)
  - — A special 4x4 control module has been adopted for the 4x4 control system control, and it operates the transfer motor according to the signals from the speed sensor, 4x4 switch, and digital TR sensor to perform shifting between 2H, 4H and 4L.
  - — The 4x4 CM receives signals from the transfer case sensor and other switches. It then activates one of two solenoids, causing a vacuum from the vacuum pump to be applied to the RFW actuator, and switching the remote freewheel (RFW) unit to lock or free.
  - — To prevent reduced performance of the actuator as a result of a decrease in engine vacuum or vacuum pump, such as during acceleration or at high altitude operation, the actuation system includes a one-way check valve.
  - — The switching mechanism of the RFW unit is a mechanical dog clutch on the left side of the front differential.
  - — The RFW operation is controlled by 4x4 CM.



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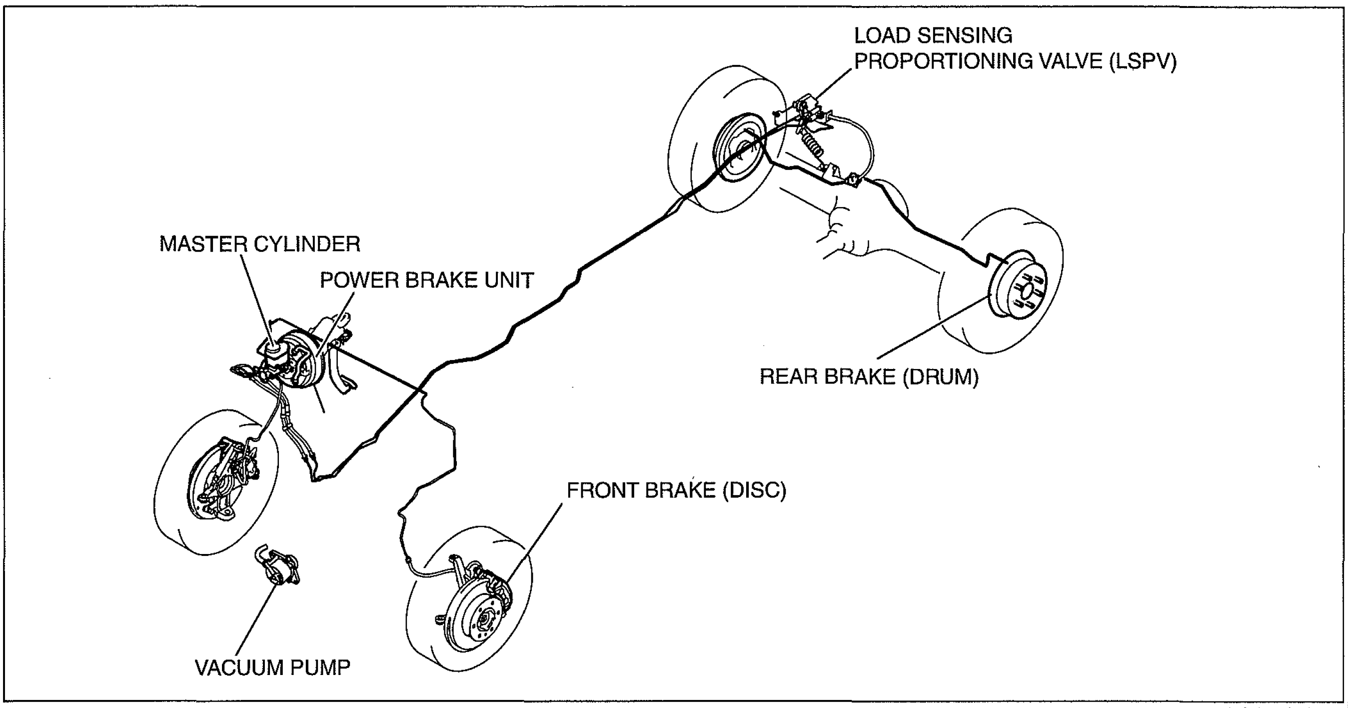
Brakes

- A tandem-type master cylinder has been adopted, improving braking force.
- A large diameter, tandem diaphragm power brake unit has been adopted, improving braking force.
- A large diameter, ventilated disc-type front brake has been adopted, improving braking force.
- A two-piston type front disc brake caliper has been adopted, improving braking force. (Hi-Rider, 4x4)
- A wide width lining for the rear brake drum has been adopted, improving braking force.
- An Automatic adjustment mechanism rear brake (drum) has been adopted, improving serviceability.
- A vacuum pump has been adopted, improving braking force.



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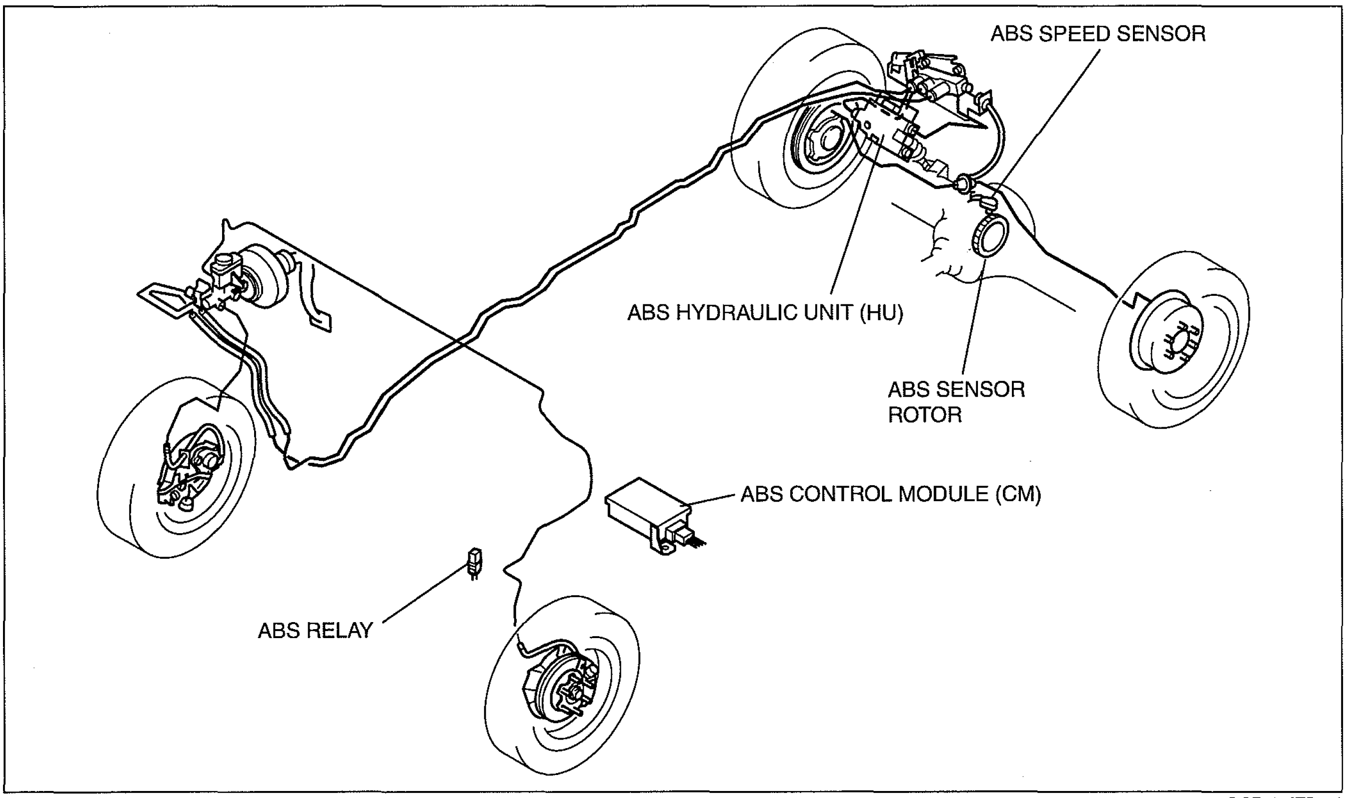
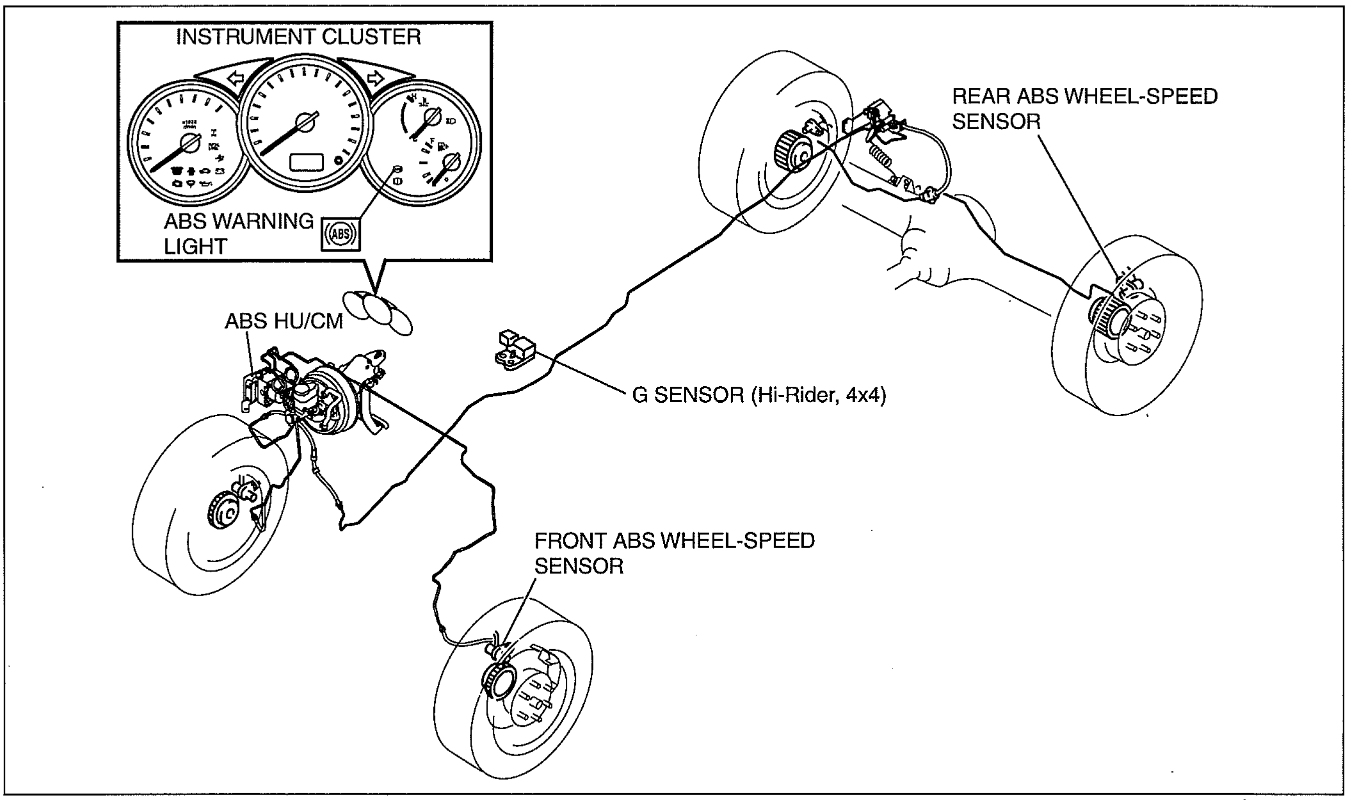
ABS [REAR ABS]

- The ABS is a one sensor, 1-channel control system, which provides dual control for the rear wheels.
- This system continuously monitors rear wheel speed via a sensor mounted on the differential. When the teeth on the sensor rotor, mounted on the ring gear, pass the sensor pickup, AC voltage is induced in the sensor circuit with a frequency proportional to the average rear wheel speed. If a rear wheel is judged likely to lockup during braking, the ABS modulates hydraulic pressure to the rear brakes, inhibiting lockup.



ABS [4W-ABS]

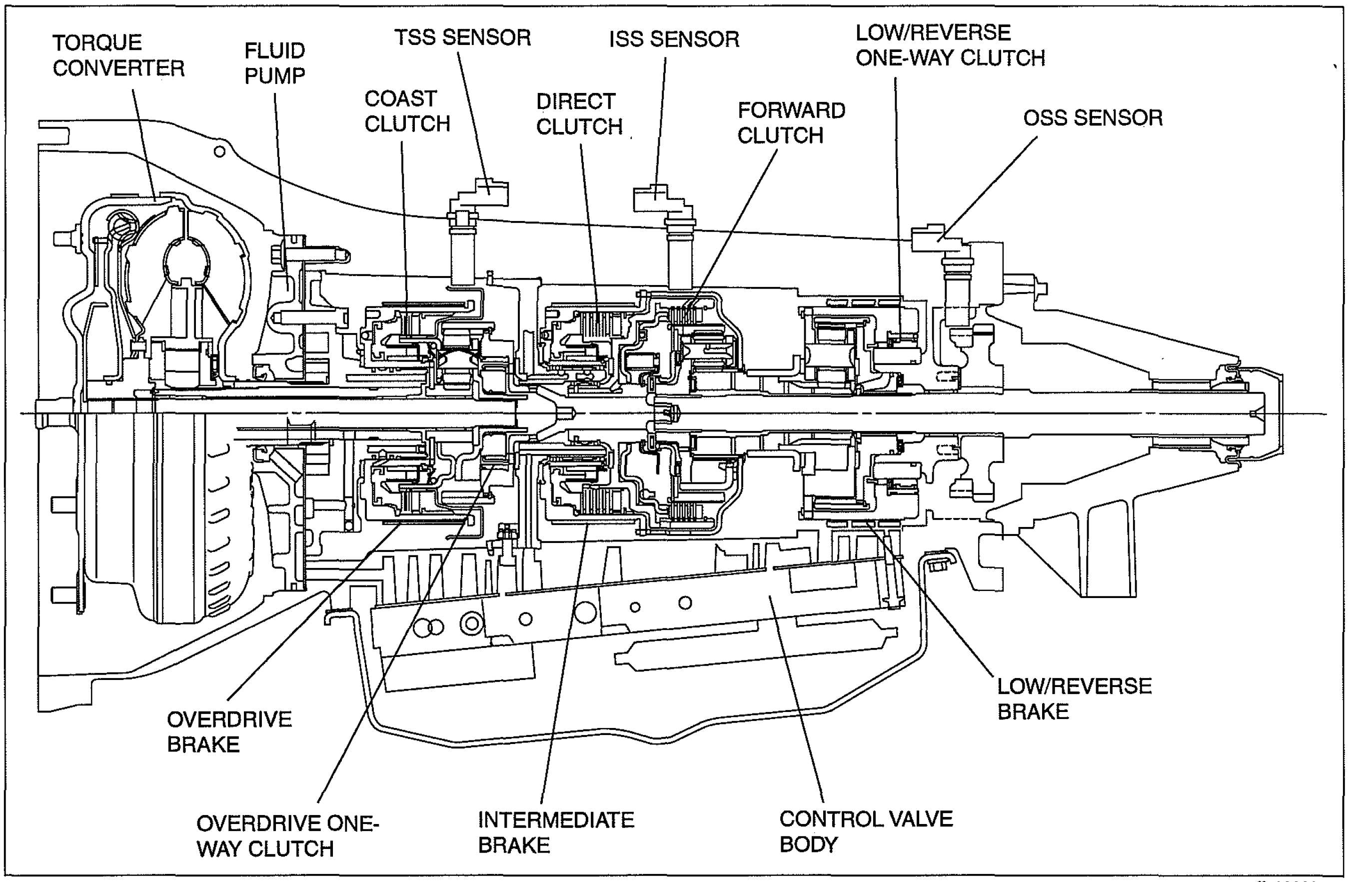
- The ABS HU/CM, integrating both the hydraulic unit (HU) and control module (CM), has been adopted, resulting in size and weight reduction.
- Select-low controlled-4-wheel anti-lock brake system with 4-sensor and 3-channel has been adopted, and has the following features.
  — The integrated ABS Hydraulic Unit/Control Module (HU/CM) system is compact, lightweight and highly reliable.



DCF413BTB001

Transmission

- Automatic transmission [5R55S]
  — Five forward speeds automatic transmission has been adopted.
  — A water-cooling type and an air-cooling type AT oil cooler has been adopted.
  — An adaptive learn strategy system has been adopted.
  — An engine-transaxle total control system has been adopted.



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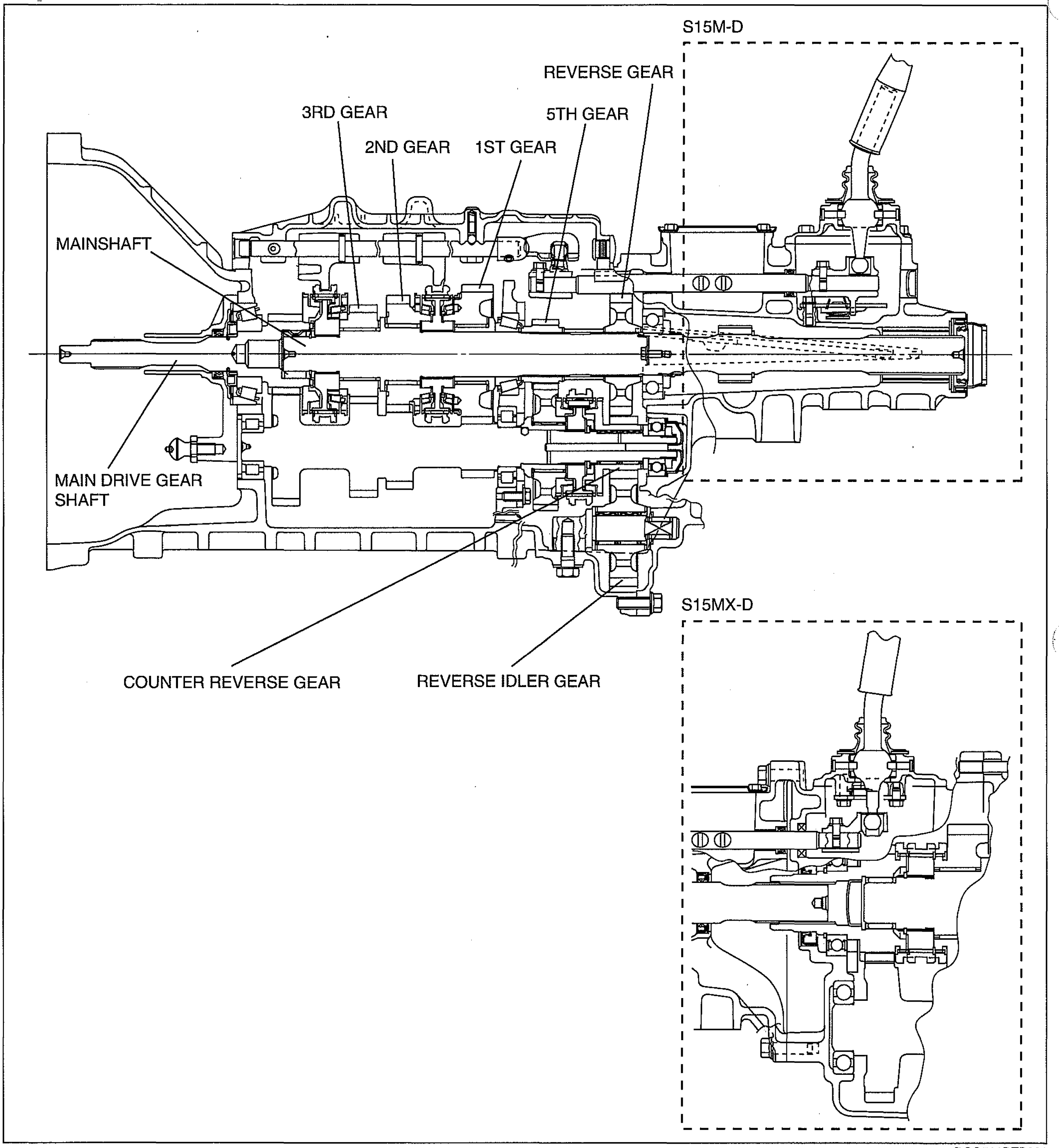
- Manual transmission [S15M-D, S15MX-D]
  — A linked, triple-cone synchronizer mechanism has been adopted for 1st and 2nd gears.
  — A linked, double cone synchronizer mechanism has been adopted for 3rd gear.
  — A cam-type reverse lockout mechanism has been adopted.

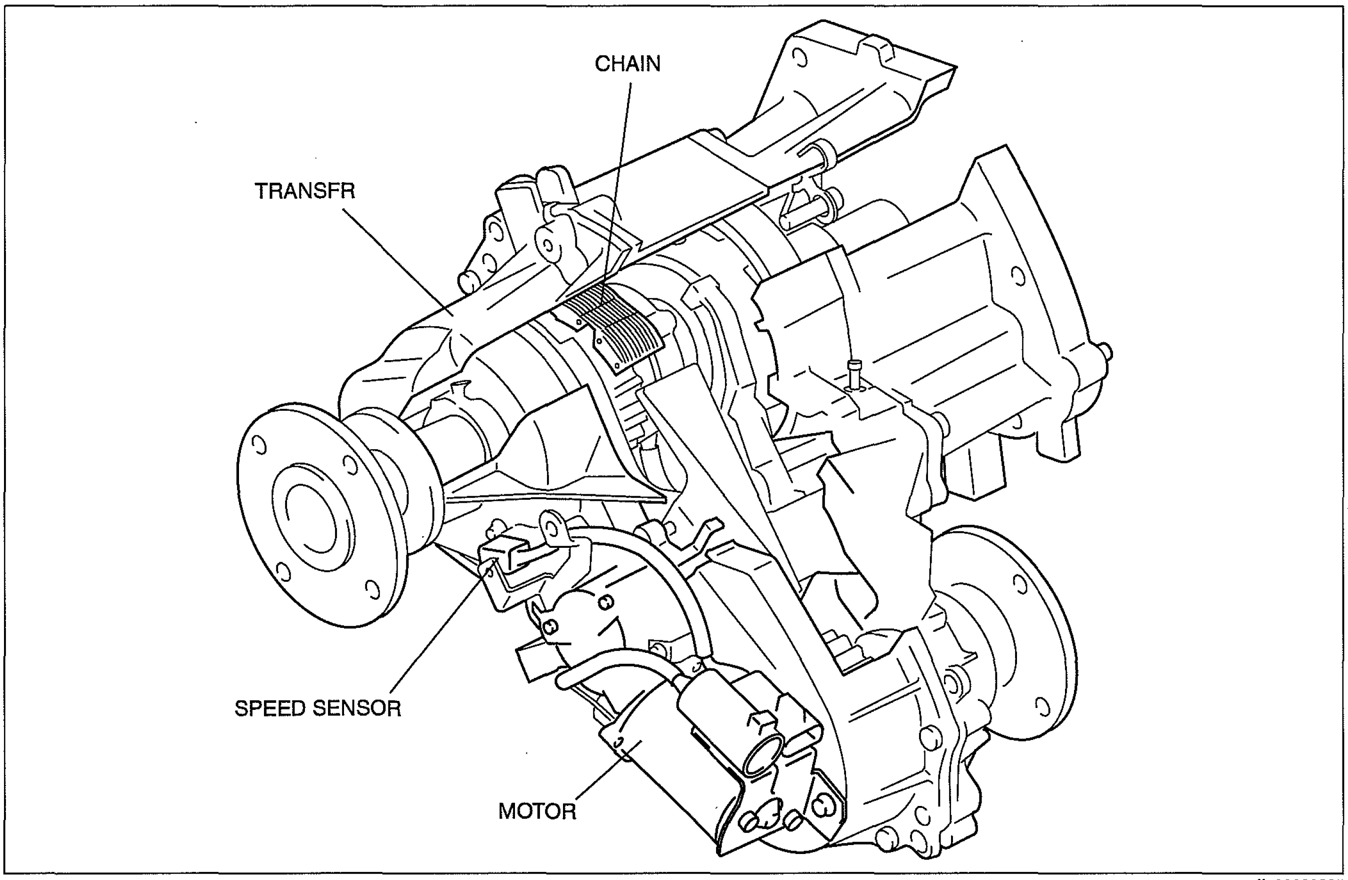


DCG511BTB001

Safety

- An immobilizer system has been adopted. This anti-theft device prevents the engine from being started unless the encrypted identification code, transmitted from a special electronic chip embedded in the key, corresponds with the identification code registered in the vehicle.
- Side air bags that effectively protect the chest area have been adopted for the seats.
- Pre-tensioner and load limiter mechanisms have been adopted for the seat belts.

- Transfer [5R55S]
  - — A chain-type transfer drive has been adopted for 4x4 driving to improve noise suppression. In addition, the drive sprocket, which drives the chain, rotates only during 4x4 driving to improve noise suppression during 4x2 driving.
  - — Shifting between 4x2 HIGH, 4x4 HIGH, and 4x4 LOW is performed by the motor for improved operability.



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VEHICLE IDENTIFICATION NUMBER (VIN) CODE

Ⓐ M N B B S 1 D 1 0 6 Ⓐ 1 2 3 4 5 6 Ⓐ

	Serial No.	
Algeria: Others:	Plant	Ⓐ mark W=A. A. Thailand
	Model Yesr Production year	7= 2007, 8=2008, 9=2009... 6= 2006, 7=2007, 8=2008...
Gulf: Algeria: Others:	Check Digit Plant No meaning	0 to 9, X W= A. A. Thailand 0
	Engine type	6= F2E (2.2 EGI) 1= WL-C (2.5 L-DI) 9= WE-C (3.0 L-DI) 2= WL-3 (2.5 L Diesel-Emission Turbo) 4= WL-Turbo (2.5L Diesel turbo) 7= G6E (2.6 EGI)
Gross vehicle weight	Gulf only 1= 2=	Other D= 2268—2721 kg {5001—6000 lbs} E= 2722—3175 kg {6001—7000 lbs}
Body style	Gulf only 8= 3= 2=	Others A= Regular cab.-without box B= Regular cab.-with box E= Double cab.-without box F= Double cab.-with box 1= Stretch cab. (with Rear Access System) -without box 2= Stretch cab. (with Rear Access System) -with box
	Product sorce	M= Thailand (for Europe (L.H.D.)) S= Japan (for Australia, General (R.H.D.), General (L.H.D.), Gulf)
	Air bag	B= Seatbelt only D= with Air bag (Driver side) L= with Air bag (Driver and Passenger) U= with Air bag (Driver, Passenger and Side air bag)
	World manufacturer identification	WF0=FORD (European (L.H.D.)) MNA=FORD (Australian) MNB=FORD (General (R.H.D.)) MNC=FORD (General (L.H.D.), Gulf)

Ⓐ : This mark is used only for Algerian model, in order to identify the manufacturer.

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Australian specs.

MNA BS******
MNA DS******
MNA LS******
MNA US******

arnffw00001612

UNITS

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Electrical current	A (ampere)
Electric power	W (watt)
Electric resistance	ohm
Electric voltage	V (volt)
Length	mm (millimeter)
	in (inch)
Negative pressure	kPa (kilo pascal)
	mmHg (millimeters of mercury)
	inHg (inches of mercury)
Positive pressure	kPa (kilo pascal)
	kgf/cm^2 (kilogram force per square centimeter)
	psi (pounds per square inch)
Torque	N·m (Newton meter)
	kgf·m (kilogram force meter)
	kgf·cm (kilogram force centimeter)
	ft·lbf (foot pound force)
	in·lbf (inch pound force)
Volume	L (liter)
	US qt (U.S. quart)
	Imp qt (Imperial quart)
	ml (milliliter)
	cc (cubic centimeter)
	cu in (cubic inch)
	fl oz (fluid ounce)
Weight	g (gram)
	oz (ounce)

Conversion to SI Units (Système International d'Unités)

- All numerical values in this manual are based on SI units. Numbers shown in conventional units are converted from these values.

Rounding Off

- Converted values are rounded off to the same number of places as the SI unit value. For example, if the SI unit value is 17.2 and the value after conversion is 37.84, the converted value will be rounded off to 37.8.

Upper and Lower Limits

- When the data indicates upper and lower limits, the converted values are rounded down if the SI unit value is an upper limit and rounded up if the SI unit value is a lower limit. Therefore, converted values for the same SI unit value may differ after conversion. For example, consider 2.7 kgf/cm^2 in the following specifications:

  210—260 kPa {2.1—2.7 kgf/cm^2, 30—38 psi}
  270—310 kPa {2.7—3.2 kgf/cm^2, 39—45 psi}

- The actual converted values for 2.7 kgf/cm^2 are 265 kPa and 38.4 psi. In the first specification, 2.7 is used as an upper limit, so the converted values are rounded down to 260 and 38. In the second specification, 2.7 is used as a lower limit, so the converted values are rounded up to 270 and 39.

NEW STANDARDS

dcf000000000t05

- Following is a comparison of the previous standard and the new standard.

New Standard		Previous Standard		Remark
Abbreviation	Name	Abbreviation	Name	
AP	Accelerator Pedal	—	Accelerator Pedal	
APP	Accelerator Pedal Position	—	Accelerator Pedal Position	
ACL	Air Cleaner	—	Air Cleaner	
A/C	Air Conditioning	—	Air Conditioning	
BARO	Barometric Pressure	—	Atmospheric Pressure	
B+	Battery Positive Voltage	V_B	Battery Voltage	
—	Brake Switch	—	Stoplight Switch	
—	Calibration Resistor	—	Corrected Resistance	#6
CMP sensor	Camshaft Position Sensor	—	Crank Angle Sensor	
LOAD	Calculated Load Voltage	—	—	
CAC	Charge Air Cooler	—	Intercooler	
CLS	Closed Loop System	—	Feedback System	
CTP	Closed Throttle Position	—	Fully Closed	
CPP	Clutch Pedal Position	—	Clutch Position	
CIS	Continuous Fuel Injection System	EGI	Electronic Gasoline Injection System	
CS sensor	Control Sleeve Sensor	CSP sensor	Control Sleeve Position Sensor	#6
CKP sensor	Crankshaft Position Sensor	—	Crank Angle Sensor 2	
DLC	Data Link Connector	—	Diagnosis Connector	
DTM	Diagnostic Test Mode	—	Test Mode	#1
DTC	Diagnostic Trouble Code(s)	—	Service Code(s)	
DI	Distributor Ignition	—	Spark Ignition	
DLI	Distributorless Ignition	—	Direct Ignition	
EI	Electronic Ignition	—	Electronic Spark Ignition	#2
ECT	Engine Coolant Temperature	—	Water Thermo	
EM	Engine Modification	—	Engine Modification	
—	Engine Speed Input Signal	—	Engine RPM Signal	
EVAP	Evaporative Emission	—	Evaporative Emission	
EGR	Exhaust Gas Recirculation	—	Exhaust Gas Recirculation	
FC	Fan Control	—	Fan Control	
FF	Flexible Fuel	—	Flexible Fuel	
4GR	Fourth Gear	—	Overdrive	
—	Fuel Pump Relay	—	Circuit Opening Relay	#3
FSO solenoid	Fuel Shut Off Solenoid	FCV	Fuel Cut Valve	#6
GEN	Generator	—	Alternator	
GND	Ground	—	Ground/Earth	
HO2S	Heated Oxygen Sensor	—	Oxygen Sensor	With heater
IAC	Idle Air Control	—	Idle Speed Control	
—	IDM Relay	—	Spill Valve Relay	#6
—	Incorrect Gear Ratio	—	—	
—	Injection Pump	FIP	Fuel Injection Pump	#6
—	Input/Turbine Speed Sensor	—	Pulse Generator	
IAT	Intake Air Temperature	—	Intake Air Thermo	
KS	Knock Sensor	—	Knock Sensor	
MIL	Malfunction Indicator Lamp	—	Malfunction Indicator Light	
MAP	Manifold Absolute Pressure	—	Intake Air Pressure	
MAF	Mass Air Flow	—	Mass Air Flow	
MAF sensor	Mass Air Flow Sensor	—	Airflow Sensor	
MFL	Multiport Fuel Injection	—	Multiport Fuel Injection	
OBD	On-Board Diagnostic	—	Diagnosis/Self Diagnosis	
OL	Open Loop	—	Open Loop	

New Standard		Previous Standard		Remark
Abbreviation	Name	Abbreviation	Name	
—	Output Speed Sensor	—	Vehicle Speed Sensor 1	
OC	Oxidation Catalytic Converter	—	Catalytic Converter	
O2S	Oxygen Sensor	—	Oxygen Sensor	
PNP	Park/Neutral Position	—	Park/Neutral Range	
PID	Parameter Identification	—	Parameter Identification	
—	PCM Control Relay	—	Main Relay	#6
PSP	Power Steering Pressure	—	Power Steering Pressure	
PCM	Powertrain Control Module	ECU	Engine Control Unit	#4
—	Pressure Control Solenoid	—	Line Pressure Solenoid Valve	
PAIR	Pulsed Secondary Air Injection	—	Secondary Air Injection System	Pulsed injection
—	Pump Speed Sensor	—	NE Sensor	#6
RAM	Random Access Memory	—	—	
AIR	Secondary Air Injection	—	Secondary Air Injection System	Injection with air pump
SAPV	Secondary Air Pulse Valve	—	Reed Valve	
SFI	Sequential Multipoint Fuel Injection	—	Sequential Fuel Injection	
—	Shift Solenoid A	—	1–2 Shift Solenoid Valve	
		—	Shift A Solenoid Valve	
—	Shift Solenoid B	—	2–3 Shift Solenoid Valve	
		—	Shift B Solenoid Valve	
—	Shift Solenoid C	—	3–4 Shift Solenoid Valve	
3GR	Third Gear	—	3rd Gear	
TWC	Three Way Catalytic Converter	—	Catalytic Converter	
TB	Throttle Body	—	Throttle Body	
TP	Throttle Position	—	—	
TP sensor	Throttle Position Sensor	—	Throttle Sensor	
TCV	Timer Control Valve	TCV	Timing Control Valve	#6
TCC	Torque Converter Clutch	—	Lockup Position	
TCM	Transmission (Transaxle) Control Module	—	EC-AT Control Unit	
—	Transmission (Transaxle) Fluid Temperature Sensor	—	ATF Thermosensor	
TR	Transmission (Transaxle) Range	—	Inhibitor Position	
TC	Turbocharger	—	Turbocharger	
VSS	Vehicle Speed Sensor	—	Vehicle Speed Sensor	
VR	Voltage Regulator	—	IC Regulator	
VAF sensor	Volume Air Flow Sensor	—	Air Flow Sensor	
WUTWC	Warm Up Three Way Catalytic Converter	—	Catalytic Converter	#5
WOT	Wide Open Throttle	—	Fully Open	

#1: Diagnostic trouble codes depend on the diagnostic test mode
#2: Controlled by the PCM
#3: In some models, there is a fuel pump relay that controls pump speed. That relay is now called the fuel pump relay (speed).
#4: Device that controls engine and powertrain
#5: Directly connected to exhaust manifold
#6: Part name of diesel engine